THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The altitude to the hypotenuse of a right triangle divides it into 5 cm and 20 cm. The altitude (cm) is: 1

(a) 10 **(b)** 25 **(c)** $\frac{25}{2}$ **(d)** 100

2. A pole 3 m high casts a shadow 2 m long. A tower casting a shadow 30 m has height (m): 1

(a) 28 **(b)** 45 **(c)** 31 **(d)** 90

3. The diagonals of a rhombus are 16 and 30. Its side squared is: 1

(a) 529 **(b)** 1156 **(c)** 578 **(d)** 289

4. Two poles of heights 17 m and 9 m are 15 m apart. The distance between their tops (m) is: 1

(a) 8 **(b)** 17 **(c)** 23 **(d)** 15

5. If $DE \parallel BC$ in $\triangle ABC$, $AB = 12$ cm, $AD = 3$ cm, $AC = 36$ cm, then EC (cm) is: 1

(a) 9 **(b)** 36 **(c)** 18 **(d)** 27

6. The areas of two similar triangles are 3 cm² and 192 cm². The ratio of their perimeters is 1: 1

(a) 8 **(b)** 9 **(c)** 64 **(d)** 16

7. The altitude to the hypotenuse of a right triangle divides it into 8 cm and 18 cm. The altitude (cm) is: 1

(a) 26 **(b)** 13 **(c)** 12 **(d)** 144

8. A pole 12 m high casts a shadow 7 m long. A tower casting a shadow 35 m has height (m): 1

(a) 40 **(b)** 60 **(c)** 28 **(d)** 420

9. The diagonals of a rhombus are 14 and 48. Its side squared is: 1

(a) 2500 **(b)** 625 **(c)** 1250 **(d)** 961

10. Two poles of heights 25 m and 5 m are 21 m apart. The distance between their tops (m) is: 1

(a) 21 **(b)** 41 **(c)** 29 **(d)** 20

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, D and E are the midpoints of AB and AC and $BC = 18$ cm. Find DE and the ratio of areas of $\triangle ADE$ and $\triangle ABC$. 2

12. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 10$ cm, $AD = 5$ cm and $AC = 20$ cm, find AE and EC. 2

13. Two similar triangles have perimeters 14 cm and 70 cm. If a side of the smaller is 2 cm, find the corresponding side of the larger and the ratio of their areas. 2

14. In $\triangle ABC$, D and E are the midpoints of AB and AC and $BC = 22$ cm. Find DE and the ratio of areas of $\triangle ADE$ and $\triangle ABC$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 13$ cm, $AD = 3$ cm and $AC = 39$ cm, find AE and EC. 3

16. $\triangle ABC \sim \triangle PQR$ with $AB = 5$ cm, $BC = 9$ cm, $CA = 12$ cm and $PQ = 15$ cm. Find QR and RP. 3

17. In a right triangle, the altitude to the hypotenuse divides it into segments of 8 cm and 18 cm. Find the altitude. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 11$ cm, $AD = 7$ cm and $AC = 22$ cm, find AE and EC. 3

18. $\triangle ABC \sim \triangle PQR$ with $AB = 7$ cm, $BC = 10$ cm, $CA = 11$ cm and $PQ = 14$ cm. Find QR and RP. 3

OR

In a right triangle, the altitude to the hypotenuse divides it into segments of 9 cm and 16 cm. Find the altitude. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two poles of heights 16 m and 4 m stand on level ground 35 m apart. Find the distance between their tops. 5

OR

The diagonals of a rhombus are 24 cm and 32 cm. Find its side. 5

20. Two poles of heights 11 m and 3 m stand on level ground 15 m apart. Find the distance between their tops. 5

OR

The diagonals of a rhombus are 14 cm and 48 cm. Find its side. 5

END OF QUESTION PAPER — Total: 40 Marks